

LFX Mentorship Coding Challenge Submission

Advanced Threshold Key Management / Lockness

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Repository: <https://github.com/parrth20/LFX-26-assignment>
Visibility: Public

Setup

Install a standard Rust toolchain with Cargo. The crate is platform independent and should compile on Linux and macOS.

Verification Commands

```
cargo fmt --check
cargo clippy --all-targets -- -D warnings
cargo test
```

The submitted code was checked locally with the commands above. Tests include the provided vectors for Ed25519, secp256k1, and secp384r1, plus round-trip tests for the implemented encryption and decryption functions.

Library API

The crate exports `encrypt(pk, message, rng)`, `decrypt(sk, ciphertext)`, and `Error`. Both functions are generic over curves supported by the `generic-ec` crate.